

Data Cleaning Interview Questions and Answers

1. What are missing values and how do you handle them?

Missing values are empty or unavailable data entries in a dataset. They can occur due to human error, system failures, or incomplete data collection.

Common ways to handle missing values include:

Removing rows or columns with too many missing values.

Filling missing values with an appropriate value such as the mean, median, mode.

Using interpolation or predictive techniques for numerical data when appropriate.

The best method depends on the amount of missing data and the importance of the affected column.

2. How do you treat duplicate records?

Duplicate records are identical rows that appear more than once in a dataset. They can lead to inaccurate analysis and misleading results.

To treat duplicate records:

Identify duplicates using tools like Excel's Remove Duplicates feature.

Verify that the duplicate records are truly unnecessary before deleting them.

3. What is the difference between `dropna()` and `fillna()` in Pandas?

`dropna()`

Removes rows or columns containing missing values and useful when only a few records contain missing data.

`fillna()`

Replaces missing values with another value, such as the mean, median and mode.

Useful when keeping all records is important.

4. What is outlier treatment and why is it important?

Outliers are data points that are significantly different from the rest of the data. They may result from errors or represent unusual observations.

Outlier treatment involves:

Identifying outliers using box plots, scatter plots, or statistical methods like the Interquartile Range (IQR) or Z-score.

Removing, transforming, or capping extreme values when appropriate.

Treating outliers is important because they can distort averages, statistical analyses, and machine learning models.

5. Explain the process of standardizing data.

Standardizing data means making the dataset consistent.

Examples include:

Changing "male", "Male", and "MALE" to "Male".

Using one date format such as DD-MM-YYYY.

Renaming column headers using lowercase letters and underscores.

Ensuring units of measurement are consistent.

Standardization improves data quality and makes analysis more accurate.

6. How do you handle inconsistent data formats (e.g., date/time)?

I first identify different formats in the dataset, then convert them into a single standard format.

For example:

2024/05/10

10-05-2024

May 10, 2024

can all be converted to:

10-05-2024

In Excel, I use date formatting tools.

7. What are common data cleaning challenges?

Some common challenges include:

Missing values

Duplicate records

Inconsistent formatting

Incorrect data types

Outliers

Inconsistent naming conventions

These issues reduce data quality and must be addressed before analysis.

8. How can you check data quality?

Data quality can be checked by:

Looking for missing values.

Identifying duplicate records.

Verifying data types.

Checking for inconsistent formats.

Detecting outliers.

Validating values against expected ranges.

Ensuring the dataset is complete, accurate, and consistent before analysis.